

Thien An NGUYEN

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Education

- USTH & INP Toulouse**, M.Sc. in Data Science Sept 2024 – Sept 2026
- **Program:** France-Vietnam Double Degree Program
 - **GPA:** 15.04/20 (French system) ~3.57/4.0 (US conversion)
 - **Coursework:** Machine Learning, Deep Learning, Modeling Techniques, Information Systems, Security and Ethics for Data, Systems Architecture, HPC Programming, Cloud and Big Data, etc.
- University of Science and Technology of Hanoi**, BSc in ICT Sept 2021 – Aug 2024
- **GPA:** 16.24/20 (French system) ~3.61/4.0 (US conversion)
 - **Coursework:** Machine Learning, Deep Learning, Web Application Development, Mobile Application Development, Digital Signals Processing, Natural Language Processing, Distributed Systems, Cryptography, Database, etc.

Achievements

- University of Science and Technology of Hanoi, Merit Scholarship 2022, 2023, 2025
- University of Science and Technology of Hanoi, Pathway to Success Scholarship 2024
- University of Science and Technology of Hanoi, Internship Scholarship 2023

Publications

Tung X. Nguyen[†], **Thien-An Nguyen[†]**, Hoang-Tung Tran. *Structure-Aware Multi-Teacher Distillation via Reinforcement Learning for Few-Shot Open-Set Keyword Spotting*.

Status: Manuscript in preparation for Interspeech 2026. (Preprint available upon request).

[†]Equal contribution

Experience

- Research Assistant**, ICT Lab - University of Science and Technology of Hanoi Feb 2025 – Present
- Working on on-device open-set KWS via incremental learning.
 - Implemented context-aware distillation loss for better distillation from large models for on-device KWS model.
 - Implemented RL for multi-teacher distillation.
- AI Engineer**, eUp Group (Full-time) Jan 2025 – Present
- Implemented RAG pipelines and vector embeddings for personalized dictionary lookup on Mazii.
 - Built multilingual explanation modules using LLMs for Japanese questions on Migii-JLPT
 - Developed AI for generating Japanese exercises that match difficulty levels. (Migii-JLPT)
 - Developing adaptive learning tracks based on user performance metrics. (Migii-JLPT)
 - Optimizing acoustic models for Japanese pronunciation scoring using speech-to-text alignment (JOPT).
 - Conducting R&D on on-device translation models from speech to speech.
 - Automating Japanese data labeling workflows using low-code agents (n8n).
- Teaching Assistant**, University of Science and Technology of Hanoi (Part-time) Oct 2024 – Mar 2025
- Teaching Course(s): Signals and Systems, NLP.
 - Prepare exercises.
 - Helped 100+ undergraduate students in Signal Processing and NLP foundation.
- Generative AI Developer**, Amela Technology (Full-time) Nov 2024 – Jan 2025
- Investigated OCR feasibility for Japanese handwriting recognition using Vision-Language Models.

- Prototyped Japanese handwriting generation pipelines using Diffusion Models.

Research Intern, L3i Lab - La Rochelle University – La Rochelle, France. (Full-time)

Apr 2024 – Jul 2024

- Investigated Computational Creativity architectures for symbolic music generation.
- Engineered split tokenization strategy to isolate pitch-velocity, and rhythm, enabling explicit modeling of musical structure.
- Implemented a GPT-2 based Transformer backbone from scratch with Relative Attention mechanisms to capture long-range dependencies.
- Evaluated objectively the performance of Symbolic Music Generation models assistively and autonomously.
- Proposed a method to evaluate the performance of the models subjectively.

Projects

Indoor Navigation Assistant for the Visually Impaired

Nov 2023 – Jan 2024

Developed a real-time computer vision system to assist visually impaired users in indoor navigation and scene understanding.

- Integrated YOLOv8 for object detection and MiDaS for monocular depth estimation to identify safe paths.
- Optimized inference pipelines on ONNX to ensure low-latency performance for real-time auditory feedback.

Edge AI Vietnamese Keyword Spotting

Feb 2023 – Aug 2023

Designed and deployed a few-shot learning framework for Vietnamese Keyword Spotting on resource-constrained edge devices.

- Research on deploying optimized Keyword Transformer models on AIoT AutoCar II and NaoV6 robots for voice-command control.
- Conducted testing on edge hardware to balance model accuracy with inference speed and power consumption.

Technologies

Programming Languages: Python, C/C++, Java, JS, PHP, HTML/CSS.

Frameworks: PyTorch, TensorFlow, Cuda jit, etc.

Technologies: Git, shell, Jupyter Notebook, GPU Clusters, etc.

Operating Systems: MacOS, Linux, Windows.

Additional Skills

Languages: Fluent in English and Vietnamese; German at B1 level; French at A2 level.

Soft skills: Independent research, self-study, teamwork, problem-solving skills.

Music Performance: Bass guitarist for NewCreation Band (NCB); 8+ years of ensemble performance experience.

Community & Church Service: Volunteer bass guitarist, serving weekly worship and community events.

References

Hoang Tung Tran, PhD - Deputy Director

Department of Information and Communication Technology, University of Science and Technology of Hanoi.

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Giang Son Tran, Assoc. Prof. - Director

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Additional: Provide when needed.